

Pilot study of dietary fat restriction and flaxseed supplementation in men with prostate cancer before surgery: exploring the effects on hormonal l...



OBJECTIVES: Dietary fat and fiber affect hormonal levels and may influence cancer progression. Flaxseed is a rich source of lignan and omega-3 fatty acids and may thwart prostate cancer. The potential effects of flaxseed may be enhanced with concomitant fat restriction. We undertook a pilot study to explore whether a flaxseed-supplemented, fat-restricted diet could affect the biomarkers of prostatic neoplasia. **METHODS:** Twenty-five patients with prostate cancer who were awaiting prostatectomy were instructed on a low-fat (20% of kilocalories or less), flaxseed-supplemented (30 g/day) diet. The baseline and follow-up levels of prostate-specific antigen, testosterone, free androgen index, and total serum cholesterol were determined. The tumors of diet-treated patients were compared with those of historic cases (matched by age, race, prostate-specific antigen level at diagnosis, and biopsy Gleason sum) with respect to apoptosis (terminal deoxynucleotidyl transferase [TdT]-mediated dUTP-biotin nick end-labeling [TUNEL]) and proliferation (MIB-1). **RESULTS:** The average duration on the diet was 34 days (range 21 to 77), during which time significant decreases were observed in total serum cholesterol (201 +/- 39 mg/dL to 174 +/- 42 mg/dL), total testosterone (422 +/- 122 ng/dL to 360 +/- 128 ng/dL), and free androgen index (36.3% +/- 18.9% to 29.3% +/- 16.8%) (all $P < 0.05$). The baseline and follow-up levels of prostate-specific antigen were 8.1 +/- 5.2 ng/mL and 8.5 +/- 7.7 ng/mL, respectively, for the entire sample ($P = 0.58$); however, among men with Gleason sums of 6 or less ($n = 19$), the PSA values were 7.1 +/- 3.9 ng/mL and 6.4 +/- 4.1 ng/mL ($P = 0.10$). The mean proliferation index was 7.4 +/- 7.8 for the historic controls versus 5.0 +/- 4.9 for the diet-treated patients ($P = 0.05$). The distribution of the apoptotic indexes differed significantly ($P = 0.01$) between groups, with most historic controls exhibiting TUNEL categorical scores of 0; diet-treated patients largely exhibited scores of 1. Both the proliferation rate and apoptosis were significantly associated with the number of days on the diet ($P = 0.049$ and $P = 0.017$, respectively). **CONCLUSIONS:** These pilot data suggest that a flaxseed-supplemented, fat-restricted diet may affect prostate cancer biology and associated biomarkers. Further study is needed to determine the benefit of this dietary regimen as either a complementary or preventive therapy.